

Relational Model Goals

Some Properties of Relational DBMS

- System controls data & operations.
- Data is of critical importance.
- Data is highly structured.

ACID Properties

Atomicity	• All-or-nothing execution
Consistency	• Always in valid state
Isolation	• Concurrent users without conflict
Durability	• Permanent data

Transactional vs. Analytical

Transactional (OLTP)

- E.g., retail sales
- Track all data – CRUD
- Frequent updates
- Flexible queries
- Do everything well

Analytical (OLAP)

- E.g., analyze sales trends
- Analyze historical data
- Periodic updates
- Predetermined queries
- Optimize for specific uses

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